

Khant Thu Aung / 105292912

6.2P (reflection)

Object Oriented Programming: Key Principles and Concepts

1. Abstraction

Abstraction means hiding the complex internal details of an object and exposing only the essential parts. It lets programmers focus on what an object does instead of how it does it.

Example:

In ShapeDrawer, we call Draw() to draw multiple types of shapes. The user doesn't need to know how the shapes are drawn and it hides the process for it.

2. Encapsulation

Encapsulation means bundling data (variables) and methods (functions) into a single unit (class), while restricting direct access to some components. It protects the integrity of the object's state.

Example:

In Shape class, it has positions(x,y) that are private. We only get the value from the public property to access the values because the actual fields are declared private.

3. Inheritance

Inheritance allows a class (subclass/child) to inherit attributes and methods from another class (superclass/parent), promoting code reuse and hierarchy modeling.

Example:

In shape drawer, we have the parent class Shape. From Shape, circles, rectangles and lines are inheriting.

4. Polymorphism

Polymorphism allows the same interface or method call to behave differently depending on the object. This includes method overriding and method overloading.

Example:

Even though the circles, rectangles and lines have Draw method, they behave differently from each others.

